



# GATE ME 90 Days Challenge

Day 3/90 · Practice Set with Detailed Solutions

SUBJECT

**Engineering  
Mathematics**

TODAY'S TOPIC

**Linear Algebra**

COUNTDOWN

**108 days to GATE**

**How to use this set:** 15 questions (8 MCQ, 7 NAT) on matrix algebra, rank, systems of linear equations, eigenvalues and eigenvectors. Try them in **40 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

**Today's free test:** 7:00 PM on our website and app, 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

## Questions

**Q1** (MCQ)

The rank of the matrix  $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$  is

- (A) 0  
(C) 2

- (B) 1  
(D) 3

**Q2** (NAT)

The determinant of  $\begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$  is \_\_\_\_\_

**Q3** (MCQ)

The system  $x + y + z = 6$ ,  $x + 2y + 3z = 10$ ,  $x + 2y + \lambda z = \mu$  has **no solution** when

- (A)  $\lambda = 3$ ,  $\mu = 10$   
(C)  $\lambda = 3$ ,  $\mu \neq 10$

- (B)  $\lambda \neq 3$ , any  $\mu$   
(D)  $\lambda = 10$ ,  $\mu = 3$

**Q4** (NAT)

For  $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$ , the trace of  $A^3$  is \_\_\_\_\_

**Q5** (MCQ)

An eigenvector of  $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  corresponding to the eigenvalue 3 is

- (A)  $[1, -1]^T$   
(C)  $[1, 2]^T$

- (B)  $[1, 1]^T$   
(D)  $[2, 1]^T$

**Q6** (NAT)

For  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ,  $A^4$  can be written as  $pA + qI$ . The value of  $p$  is \_\_\_\_\_

**Q7** (MCQ)

Which statement about a real symmetric matrix is **NOT** always true?

- (A) All its eigenvalues are real  
(B) Eigenvectors for distinct eigenvalues are orthogonal  
(C) It is invertible  
(D) It can be diagonalised by an orthogonal matrix

**Q8** (NAT)

$A$  is a  $4 \times 5$  real matrix of rank 3. The number of linearly independent solutions of  $Ax = 0$  is \_\_\_\_\_

**Q9** (MCQ)

If  $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ , the eigenvalues of  $A^2 + 2A + I$  are

- (A) 3 and -1  
(B) 9 and 1  
(C) 16 and 0  
(D) 4 and 0

**Q10** (NAT)

$A$   $3 \times 3$  matrix  $A$  has eigenvalues 1, 2 and 3. The determinant of  $\text{adj}(A)$  is \_\_\_\_\_

**Q11** (MCQ)

Solving  $2x + y - z = 1$ ,  $x + 3y + 2z = 13$ ,  $3x - y + 4z = 13$  by Gauss elimination, the value of  $z$  is

- (A) 1  
(B) 2  
(C) 3  
(D) 4

**Q12** (NAT)

The rank of  $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 5 & 7 & 9 \\ 1 & 1 & 1 & 1 \end{bmatrix}$  is \_\_\_\_\_

**Q13** (MCQ)

The system  $kx + y = 0$ ,  $x + ky = 0$  has a non-trivial solution when

- (A)  $k = 0$  only  
(B)  $k = \pm 1$   
(C)  $k = \pm 2$   
(D) no real  $k$

**Q14** (NAT)

The largest eigenvalue of  $\begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$  is \_\_\_\_\_

**Q15** MCQ

The matrix  $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  has

- (A) two linearly independent eigenvectors      (B) only one linearly independent eigenvector  
(C) no real eigenvalue      (D) eigenvalues 1 and 0
-

## Answer Key

| Q   | Type | Answer                                        |
|-----|------|-----------------------------------------------|
| Q1  | MCQ  | (C) 2                                         |
| Q2  | NAT  | 41                                            |
| Q3  | MCQ  | (C) $\lambda = 3, \mu \neq 10$                |
| Q4  | NAT  | 251                                           |
| Q5  | MCQ  | (B) $[1, 1]^T$                                |
| Q6  | NAT  | 145                                           |
| Q7  | MCQ  | (C) It is invertible                          |
| Q8  | NAT  | 2                                             |
| Q9  | MCQ  | (C) 16 and 0                                  |
| Q10 | NAT  | 36                                            |
| Q11 | MCQ  | (C) 3                                         |
| Q12 | NAT  | 2                                             |
| Q13 | MCQ  | (B) $k = \pm 1$                               |
| Q14 | NAT  | 5                                             |
| Q15 | MCQ  | (B) only one linearly independent eigenvector |

## Detailed Solutions

### Q1 Answer: (C) 2

**Concept:** Rank = number of non-zero rows after row reduction = order of the largest non-zero minor.

$R_2 = 2R_1$ , so  $R_2 \rightarrow R_2 - 2R_1$  gives a zero row.  $\det A = 0$ , so rank  $< 3$ .

$R_1$  and  $R_3$  are not proportional: the minor  $\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 - 2 = -1 \neq 0$ .

Rank = 2

**Exam tip:** Look for proportional rows or columns first; it often settles the rank without full elimination.

### Q2 Answer: 41

**Concept:** Expand along a row or column with zeros to save work.

Along row 1:  $2(4 \cdot 6 - 5 \cdot 0) - 1(0 \cdot 6 - 5 \cdot 1) + 3(0 \cdot 0 - 4 \cdot 1)$

$= 2(24) - 1(-5) + 3(-4) = 48 + 5 - 12 = 41$

**Exam tip:** Watch the sign pattern  $+ - +$  along the first row; the middle term flips sign.

**Q3 Answer: (C)  $\lambda = 3, \mu \neq 10$** 

**Concept:** Compare  $\text{rank}(A)$  with rank of the augmented matrix  $[A | b]$ . Unequal ranks  $\rightarrow$  no solution.

$R_3 - R_2$  gives  $(0, 0, \lambda - 3 | \mu - 10)$ .

$\lambda \neq 3$ :  $\text{rank } A = 3 = \text{rank } [A|b] \rightarrow$  unique solution.

$\lambda = 3, \mu = 10$ : both ranks 2  $\rightarrow$  infinitely many solutions.

$\lambda = 3, \mu \neq 10$ :  $\text{rank } A = 2, \text{rank } [A|b] = 3 \rightarrow$  **no solution**.

**Exam tip:** Reduce only the row that contains  $\lambda$  and  $\mu$ ; the other rows need no work.

**Q4 Answer: 251**

**Concept:** If  $\lambda$  is an eigenvalue of  $A$  then  $\lambda^3$  is an eigenvalue of  $A^3$ , and  $\text{trace} = \text{sum of eigenvalues}$ .

$A$  is lower triangular, so its eigenvalues are the diagonal entries 2, 3, 6.

Eigenvalues of  $A^3$ : 8, 27, 216

$\text{trace}(A^3) = 8 + 27 + 216 = \mathbf{251}$

**Exam tip:** Never multiply out  $A^3$  in the exam. Triangular  $\rightarrow$  read eigenvalues from the diagonal.

**Q5 Answer: (B)  $[1, 1]^T$** 

**Concept:** Solve  $(A - \lambda I)x = 0$  for  $\lambda = 3$ .

$A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \rightarrow -x_1 + x_2 = 0 \rightarrow x_1 = x_2$

Eigenvector:  $[1, 1]^T$  (any non-zero multiple).

Check:  $A[1, 1]^T = [3, 3]^T = 3[1, 1]^T \checkmark$ .  $[1, -1]^T$  belongs to  $\lambda = 1$ .

**Exam tip:** In an MCQ, just multiply each option by  $A$  and see which comes out as  $\lambda$  times itself.

**Q6 Answer: 145**

**Concept:** Cayley-Hamilton: every square matrix satisfies its own characteristic equation.

Characteristic equation:  $\lambda^2 - (\text{trace})\lambda + \det = \lambda^2 - 5\lambda - 2 = 0 \rightarrow A^2 = 5A + 2I$

$A^3 = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I$

$A^4 = 27A^2 + 10A = 27(5A + 2I) + 10A = 145A + 54I$

$p = \mathbf{145}$  (and  $q = 54$ )

**Exam tip:** Cayley-Hamilton also gives the inverse:  $A^{-1} = (A - 5I)/2$  here.

**Q7 Answer: (C) It is invertible**

**Concept:** The spectral theorem guarantees real eigenvalues, orthogonal eigenvectors and orthogonal diagonalisation for every real symmetric matrix.

Invertibility needs every eigenvalue to be non-zero, which is not guaranteed.

Counter-example:

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  is symmetric but  $\det = 0$  (eigenvalues 0 and 2).

So '**It is invertible**' is the false statement.

**Exam tip:** Symmetric says nothing about singularity; a zero eigenvalue is allowed.

**Q8 Answer: 2**

**Concept:** Rank-nullity theorem: rank + nullity = number of unknowns (columns).

Unknowns  $n = 5$ , rank  $r = 3$

Nullity =  $n - r = 5 - 3 = 2$

**Exam tip:** Use the number of columns, not rows.  $4 - 3 = 1$  is the usual slip.

**Q9 Answer: (C) 16 and 0**

**Concept:** If  $Ax = \lambda x$  then  $f(A)x = f(\lambda)x$  for any polynomial  $f$ .

Eigenvalues of  $A$ :  $\lambda^2 - 2\lambda - 3 = 0 \rightarrow \lambda = 3, -1$

$f(\lambda) = \lambda^2 + 2\lambda + 1 = (\lambda + 1)^2 \rightarrow (3 + 1)^2 = 16$  and  $(-1 + 1)^2 = 0$

Eigenvalues of  $A^2 + 2A + I$ : **16 and 0**

**Exam tip:** Note that  $A^2 + 2A + I$  is singular here, because one of its eigenvalues is zero.

**Q10 Answer: 36**

**Concept:**  $\det(\text{adj } A) = (\det A)^{n-1}$  for an  $n \times n$  matrix.

$\det A = \text{product of eigenvalues} = 1 \times 2 \times 3 = 6$

$\det(\text{adj } A) = 6^{3-1} = 6^2 = 36$

**Exam tip:** Also remember  $\text{adj}(\text{adj } A) = (\det A)^{n-2} A$ .

**Q11 Answer: (C) 3**

**Concept:** Eliminate  $x$ , then  $y$ , then back-substitute.

$R_2 \rightarrow 2R_2 - R_1: 5y + 5z = 25 \rightarrow y + z = 5$

$R_3 \rightarrow 2R_3 - 3R_1: -5y + 11z = 23$

Adding  $5(y + z = 5): 16z = 48 \rightarrow z = 3$ , then  $y = 2$  and  $x = 1$ .

**Exam tip:** Substitute (1, 2, 3) back into all three equations. It takes 10 seconds and catches sign errors.

**Q12 Answer: 2**

**Concept:** Rank = number of linearly independent rows.

$R_2 = 2R_1$ , so  $R_2$  adds nothing.

$R_3 = 2R_1 + R_4: 2[1, 2, 3, 4] + [1, 1, 1, 1] = [3, 5, 7, 9] \checkmark$ , so  $R_3$  adds nothing either.

$R_1$  and  $R_4$  are not proportional  $\rightarrow \text{rank} = 2$

**Exam tip:** Rows whose entries are in arithmetic progression (1 2 3 4, 3 5 7 9, 1 1 1 1) always give rank  $\leq 2$ .

**Q13 Answer: (B)  $k = \pm 1$** 

**Concept:** A homogeneous system has non-trivial solutions only if the coefficient matrix is singular.

$\det = k^2 - 1 = 0 \rightarrow k = \pm 1$

Check  $k = 1: x + y = 0$  twice  $\rightarrow x = -y$ , infinitely many solutions  $\checkmark$

**Exam tip:** Homogeneous systems are always consistent; the only question is whether  $\det = 0$ .

**Q14 Answer: 5**

**Concept:**  $A = 2I + J$ , where  $J$  is the all-ones matrix.  $J$  has eigenvalues 3, 0, 0.

So  $A$  has eigenvalues  $2 + 3 = 5$ ,  $2 + 0 = 2$ ,  $2 + 0 = 2$ .

Largest = **5**. Check: trace =  $9 = 5 + 2 + 2$  ✓; det =  $20 = 5 \times 2 \times 2$  ✓

**Exam tip:** Equal row sums  $s$  mean  $s$  is an eigenvalue with eigenvector  $[1, 1, 1]^T$ . Here every row sums to 5.

**Q15 Answer: (B) only one linearly independent eigenvector**

**Concept:** Number of independent eigenvectors for  $\lambda = n - \text{rank}(A - \lambda I)$  (geometric multiplicity).

Eigenvalue  $\lambda = 1$  is repeated (algebraic multiplicity 2).

$A - I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$  has rank 1  $\rightarrow$  independent eigenvectors =  $2 - 1 = 1$  (only  $[1, 0]^T$ ).

So  $A$  is not diagonalisable.

**Exam tip:** A repeated eigenvalue does not guarantee a full set of eigenvectors; check the rank of  $A - \lambda I$ .

## What next?

### Free Day 3 test tonight, 7:00 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with photo, name and marks in the next 8 AM challenge reel.



Website test series



Dig Career Thrust app



Telegram: daily PDFs

**Next, Day 4/90:** Engineering Mechanics, Impulse, Energy & Impact. Comment **GATE90** on our reel to get all links.

Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.